DAV PUBLIC SCHOOL BSEB COLONY PATNA
HOLIDAY H.W

CLASSX.

PHYSICS REFLECTION AND REFRACTION OF LIGHT

I.CHOOSE THE CORRECT OPTION:

1. A real image is formed by the light rays after reflection or refraction when they

A. actually meet or intersect with each other

B. actually converge at a point

C appear to meet when they are produced in the backward direction

D. appear to diverge from a point

which of the above statements are correct

a. A and D

b. B and D

c. A and B

d. B and C

2. A ray of light entered medium P from medium Q its velocity increased what can be said about the refractive index of medium P as compared to that of medium Q

a. it is lower

b. it is higher

c. it is the same

d. nothing can be said

3. the power of a lens of focal length 25 centimetre in diopter is

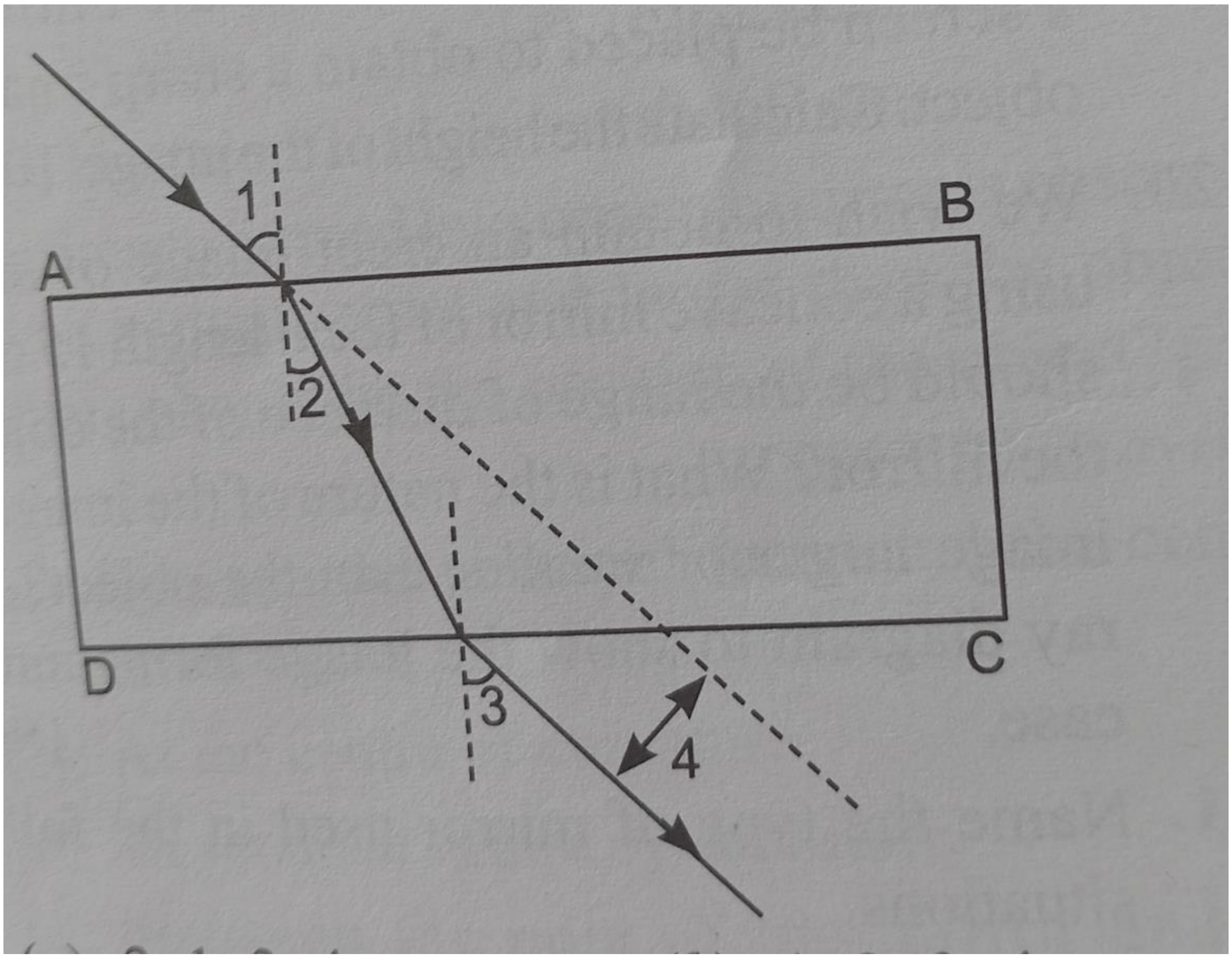
a. 0.25

b. 2.5

c. 4

d. 0.4

4. A student has traced the path of a ray of light through a glass slab as follows if you are asked to label 1,2,3,4 the correct sequence of labelling angle i,e,r and lateral displacement respectively is



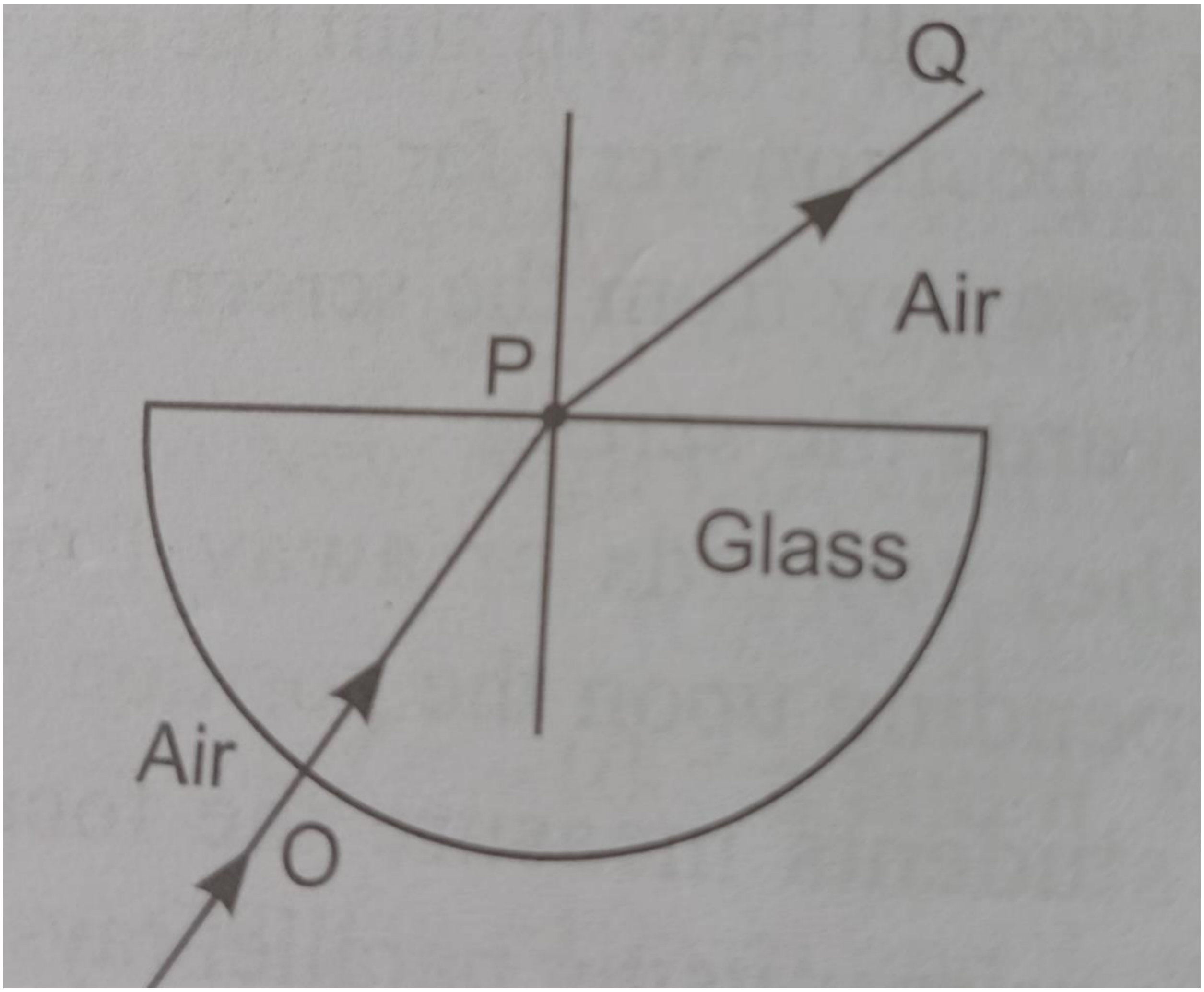
a. 2,1,3 ,4

b. 1,2,3,4

c.1,3,2,4

d.1,3,4,2

5. The angle of incidence from air to glass at a point O on the hemispherical glass slab is



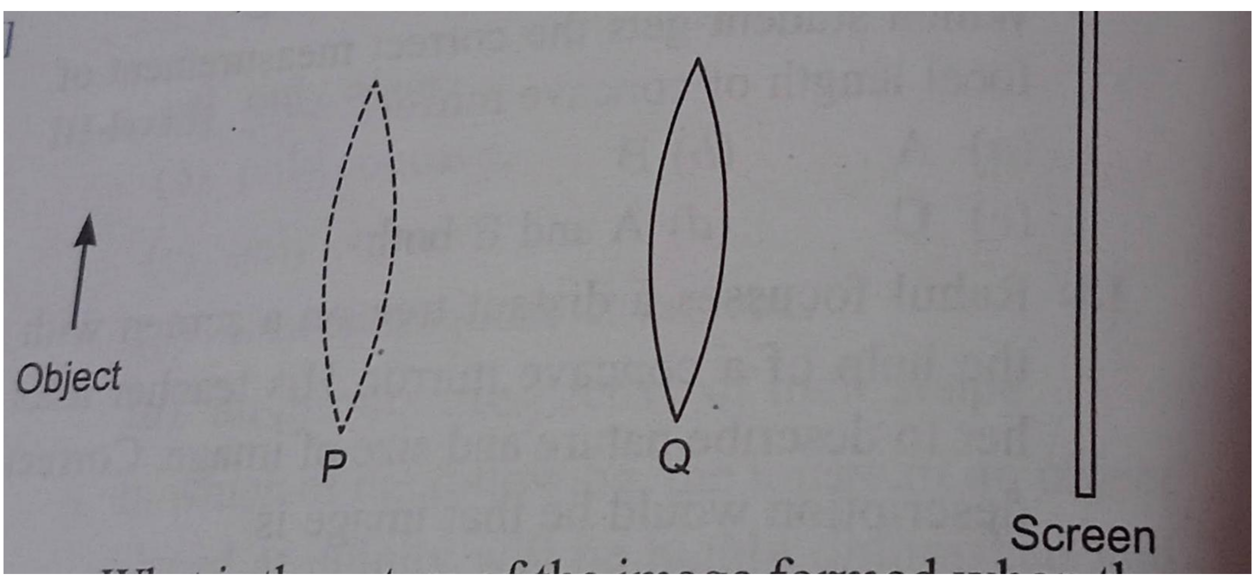
a. 45 degree

b. 0 degree

c. 90 degree

d. 180 degree

6. When a lens is placed at Q a sharp image is formed on the screen the image formed is real inverted and diminished. when the lens is moved to P another sharp images formed on the screen.



What is the nature of the image formed when the lens is at P?

a. Magnified and inverted
b. Magnified and upright
c. Diminished and upright
d. Diminished and inverted

7. If the power of lens is -4 diopter then it means that the lens is a

a. concave lens of focal length -50 metre

b convex lens of focal length -50 cm

c. concave lens of focal length - 25 cm

d. convex lens of focal length -25 cm

8.Two convex lens P and Q have focal length 0.50 metre and 0.40 metre respectively.

which of the following is TRUE about the combined power of the two lenses.

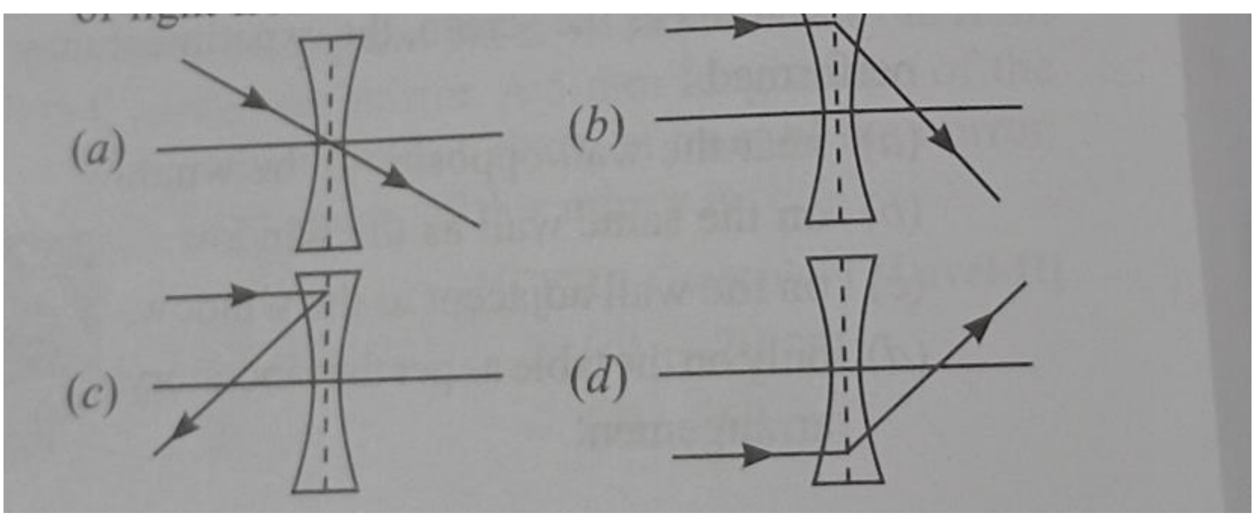
a. P is equal to 4.5 dioptre

b. P is less than 4.5 diopter

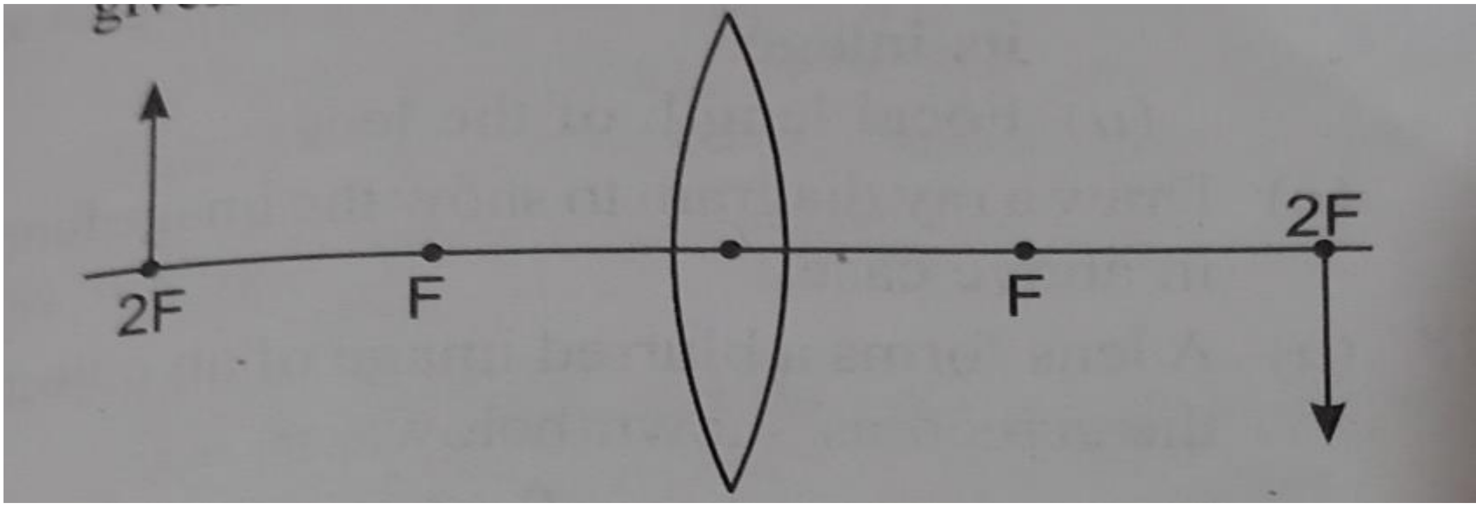
c. P is more than 4.5 diopter

d. P cannot be determined from the information given.

9. Which of the following correctly shows refraction of light from a concave lens?



10. For the diagram shown according to new cartesian sign convention the sign of object distance, image distance ,and focal length for the given lens will be



a. -,+,+
b. -,-,+
c. -,-,-
d. +,-,+

11. Where should an object is placed in front of a convex lens to get a real image of the size of the object

a. at the principal focus of the lens

b. at twice the focal length

c. at Infinity

d. between the optical centre of the lens and its principal focus

12. A spherical mirror and a thin spherical lens have each a focal length of -15 centimetre the mirror and the lens are likely to be

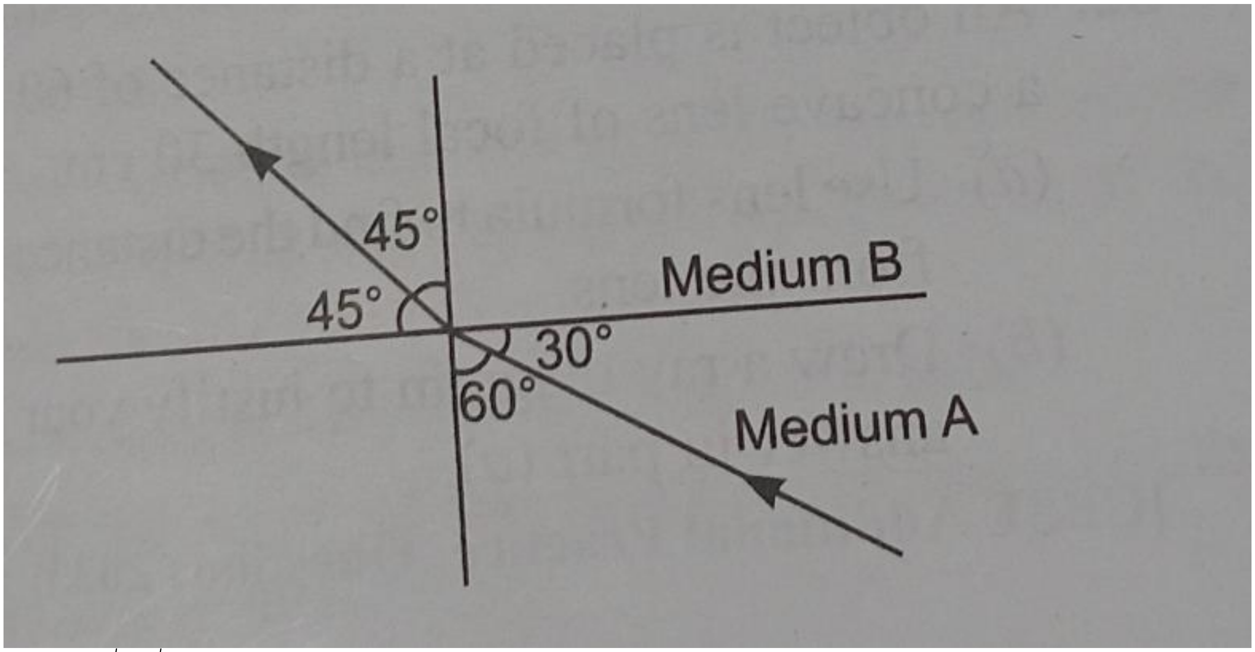
a. both concave

b. both convex

c. the mirror is concave and the lens is convex

d. the mirror is convex but the lens is concave

13. Figure shows a ray of light as it travels from medium A to medium B .refractive index of the medium b relative to medium a is



a. √3/√2
b. √2/√3
c. 1/√2
d. √2

14. A sharp image of a distant object is obtained on a screen by using a convex lens in order to determine the focal length of the lens, you need to measure the distance between the

a. lens and the object

b. lens and the screen

c. object and the screen

d. lens and the screen and also object and the screen

15. A student obtained a sharp image of the grill of a window on a screen using a convex lens for getting better results the teacher suggested focusing of a distant tree instead of the grill in which direction should the lens be moved for this purpose

a. away from the screen

b. very far away from the stream

c. behind the screen

d. towards the screen

16. A student has to do the experiment on finding the focal length of a given convex lens by using a distant object she can do her experiment if she is also made available with

a. a lamp and screen

b. a scale and the screen

c. A lamp and a scale

d. Only a screen

17. in an experiment to determine the focal length of a convex lens a student obtained a sharp and inverted image of a distant tree on the screen behind the lens she then remove the screen and looked through the lens in the direction of object she will see

a. an inverted image of the tree at the focus of the lens

b. no image as the screen has been removed

c. a blurred image on the wall of the laboratory

d. an erect image of the tree

18. An experiment to trace the path of a ray of light through a glass slab was performed by four students I ,II,III,IV . They reported the following measurements of angle of incidence i, angle of refraction r ,and angle of emergence e. which one of the students has performed the experiment correctly?

Student	Angle i	Angle r	Angle e
I	60°	35°	59°
II	45°	40°	40°
III	35°	30°	40°
IV	50°	55°	50°

a. I b. II c. III d. IV

19. In an experiment to trace the path of a ray of light passing through a rectangular glass slab students must take the precautions:

a. Range of incident angle should be 30 degrees to 60 degrees

b. The eye must be in the line with the feet of pins.

c. the pins used must have sharp point and stand vertically on the board

d. all of these

II. VERY SHORT ANSWER QUESTION

20. Explain with the help of a diagram why a pencil partly immersed in water appears to be bent at the water surface

21. A ray of light falls making an angle of incidence theta on the surface of a glass slab. Draw a labelled ray diagram to show its path also mark lateral displacement on it

22. Refractive index of diamond with respect to glass is 1.6 and absolute refractive index of glass is 1.5 find out the absolute refractive index of diamond.

23. The absolute refractive indices of glass and water are 3/2 and 4/3 respectively .if the speed of light in glass is 2x10^8 m/s , calculate the speed of light in(a) vacuum and in (b) water.

24. A real image 2/3 rd of the size of the object is formed by a convex lens when the object is at a distance of 12 cm from it. Find the focal length of the lens.

25. You are given kerosene turpentine and water in which of these does the light travel faster. why?(RI of turpentine oil =1.47 and RI of water = 1.33)

26. A ray of light travelling in air enters obliquely into water .Does the light ray bent towards the normal or away from the normal? Why?

III. SHORT ANSWER QUESTION

27.Draw a ray diagram in each of the following cases to show the formation of image when the object is placed

I . Between optical centre and principal focus of a convex lens

ii.. anywhere in front of a concave lens

iii. at 2 F of convex lens

state the signs and values of magnifications in the above mentioned cases i and ii

28. State the laws of refraction of light explain the term ‘absolute refractive index’ of a medium and write an expression to relate it with the speed of light in vacuum.

29. Draw a ray diagram to show the refraction of light through a glass slab and mark angle of refraction and the lateral shift suffered by the ray of light while passing through the slab.

30.If the refractive index of glass for light going from air to glass is 3/2 find the refractive index of air for light going from glass to air.

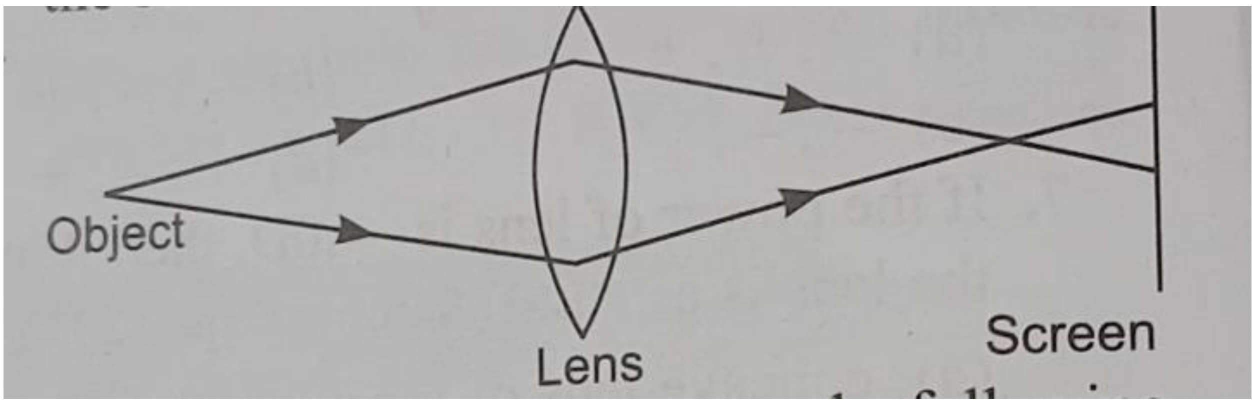
31. a. The image of an object formed by a lens is of same size but inverted if the object distance is 30 centimetre calculate

i. the distance between the object and its image

ii. focal length of the lens

b. draw a ray diagram to show the image formed in the above case

32. A lens forms a blurred image of an object on the screen as shown below



what changes can you make to the following to form a sharp and in-focus image on the screen?

i. object distance

ii. focal length of the lens

33. If the image formed by a lens for all position of an object placed in front of it is always erect and diminished what is the nature of this lens? draw a ray diagram to justify your answer if the numerical value of the power of this lens is 10 diopter what is its focal length in the cartesian system?

34. A 4 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. the distance of the object from the lens is 15 cm .find the nature position and size of the image formed.

35. An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm.

a use lens formula to find the distance of image from the lens.

b. Draw a ray diagram to justify your answer in part a.

LONG ANSWER TYPE QUESTION

36. analyse the following observation table showing variation of image distance' v'with object distance' u'in case of a convex lens and answer the questions that follows without doing any calculation.

S.no	Image distance u (cm)	Image distance v (cm)
1	-100	+25
2.	-60	+30
3.	-40	+40
4.	-30	+60
5.	-25	+100
6.	-15	+120

a. What is the focal length of the convex lens ?Give reason to justify your answer.

b. Write the serial number of observation which is not correct. On what basis have you arrived at this conclusion.

c. Select an appropriate scale and draw a ray diagram for observation at S.no 2. Also find the approximate value of magnification.

37. What is meant by the power of a lens? what is its SI unit ?Name the type of lens whose power is positive.

The image of an object formed by a lens is real inverted and of same size as the object. If the image is at a distance of 40 cm from the lens, what is the nature and power of the lens ? Draw ray diagram to justify your answer.

38. a. How can we differentiate between convex and concave lenses without touching them.

b. Two thin lenses of power + 3.5 diopter and -2.5 diopter are placed in contact .find the power and focal length of the lens combination.

c. Distinguish between a real and virtual image.

39. a. i. Draw a ray diagram to show the path of the refracted ray in each of the following cases:

A ray of light incident on a concave lens

1.parallel to its principal axis and

2. is directed towards its principle focus.

ii. A 4 cm tall object is placed perpendicular to the principal axis of convex lens of focal length 24 cm. The distance of object from the lens is 16 cm. find the position and size of the image formed.

40. A spherical lens Produces an image of magnification -1 on a screen placed at a distance of 50 cm from the lens.

a. Write the type of lens

b. find the distance of image from the lens

c. what is the focal length of the lens

d. Draw the ray diagram to show the image formation in this case.

41. Rishi went to a palmist to show his palm. the palmist use a special lens for this purpose.

i. state the nature of the lens and the reason for its use,

ii. Where should the palmist place or hold the lens so as to have a real and magnified image of an object?

iii. If the focal length of this lens is 10 cm and the lens is held at a distance of 5 cm from the palm, use lens formula to find the position and size of the image.

42. The image of an object formed by a lens is of magnification -1. if the distance between the object and its image is 60 cm, i. What is the focal length of the lens? if the object is moved 20 cm towards the lens,

ii. Where should the image be formed?

iii. State reason and also draw a re diagram in support of your answer.

CASE BASED OR SOURCE BASED QUESTION

43. A compound microscope is an instrument which consists of two lenses L1 and L2 .The lens L1 called objective ,forms a real ,inverted and magnified image of the given object. This serves as the object for the second lens L2: the eyepiece. The eyepiece functions like a simple microscope or magnifier. it produces the final image, which is inverted with respect to the original object, enlarged and virtual.

a. What type of lenses must be L1and L2 ?

b. What is the value and sign of magnification of the image formed by L1.

c. If the power of eye piece L2 is 5 diopter and it forms an image at a distance of 80 cm from its optical centre ,at what distance should the object be?

Or

If the power of lenses L1 and L2 are in the ratio of 4 :1, what would be ratio of the focal length of L1and L2.

44. a. State Snell's law of refraction. State 2 factors affecting lateral displacement.

b. What changes are observed in image formed by a convex lens whose half portion is covered with black paper . justify your answer

45. A student focused the image of a candle flame on a white screen by placing the flame at various distances from a convex lens .He noted his observation as:

S.no	Distance of flame from the lens(cm)	Distance of screen from the lens(cm)
1	60	20
2	40	24
3	30	30
4	24	40
5	15	70

a. In which case the size of the object and image will be same.
b. What is the change in the image observed as the object is moved from infinity towards the concave lens?
c. From the above table ,find the focal length of the lens without using lens formula.

OR

Which set of observation Is incorrect?

46. slides are small transparencies mounted in sturdy frames ideally suited to magnification and projection, since they have a very high resolution and a high image quality. There is a tray where the slides are to be put into a particular orientation so that the viewers can see the enlarged erect images of the transparent slides. This means that the slides will have to be inserted upside down in the projector tray. the above was the description of slide projector.

To show her students the images of insects that she investigated in the lab, Mrs. Iyer brought a slide projector. her slide projector produces a 500 times enlarged and inverted image of a slide on a screen 10 m away.

i. Based on the text and the data given in the above paragraph, what kind of lens must the slide projector have?

ii. if v is the symbol used for image distance and u for object distance then with one reason state what will be the sign for v /u in the given case?

iii. A slide projector has a convex lens with a focal length of 20 cm. the slide is placed upside down 21 cm from the lens. How far away should the screen be placed from the slide projector's lens so that the slide is in focus?

OR

When a slide is placed 15 cm behind the lens in the projector, an image is formed 3 m in front of the lens. If the focal length of the lens is 14 cm, Draw a ray diagram to show image formation.

ASSERTION AND REASON QUESTIONS

Read the statements carefully and choose the correct alternative from the following

A .both the assertion and the reason are correct and the reason is the correct explanation of the assertion.

B. The assertion and the reason are correct but the reason is not the correct explanation of the assertion.

C .Assertion is true but the reason is false.

D. The statement of the assertion is false but the reason is true.

47. Assertion: a concave lens of very short focal length causes higher divergence than one with longer focal length.

Reason the power of lens is directly proportional to its focal length

48. Assertion: the SI unit of power of lens is diopter.

Reason: the power of concave lens is positive and that of convex lens is negative

49. Assertion: when a pencil is partly immersed in water and held obliquely to the surface the pencil appears to bend at the water surface.

Reason: the apparent bending of the pencil is due to the refraction of light when it passes from water to air.

50. Assertion: Higher the refractive index of the medium, lesser will be the velocity of light in that media.

Reason: Refractive index is inversely proportional to the velocity of light in the medium.

51.Assertion : a ray of light travels from air to glass the light ray bends away from the normal.

Reason: light travels with different speeds in different medium.

52. Assertion: The bottom of a tank or pond filled with water appears to be raised.

Reason: the apparent depth of a tank is given by 1/n times the original depth.

53. Assertion: A concave lens makes an object look nearer and magnified.

Reason: concave lens always forms erect images.

54. Assertion: Light does not travel in the same direction in all the media.

Reason: the speed of light does not change as it enters from one transparent medium to another.

55. Assertion :The emergent ray is parallel to the direction of incident ray.

Reason :The extent of bending of the ray of light at the opposite parallel faces(air glass interface and glass air interface)of the rectangular glass slab is equal and opposite.

56. Assertion :A ray of light travelling from rarer medium to a denser medium slows down and bends away from the normal. when it travels from a denser medium to rarer medium, its speeds up and bends towards the normal.

Reason :the speed of light is higher in a rarer medium than a denser medium.

57. Assertion : light travels faster in glass than in air.

Reason :glass is denser than air.

58. Assertion : the height of an object is always considered positive.

Reason :an object is always placed above the principal axis in the upward direction.

59. Assertion: refractive index has no units.

Reason :the refractive index is the ratio of two similar quantities.

60. Assertion: Light ray does not deviate while passing through optical centre of a lens.

Reason: As thin lens is a combination of glass slab and prism so at the centre the slab is of negligible thickness.